

**Department of Finance
Olin Business School
Washington University in St. Louis
Research Methods in Finance**

Stata Group Assignment 1 – Group 10

1. Download monthly return data for decile portfolios formed from a characteristic(s) from **Ken French's Data Library**. (Please do not use the decile portfolios from the in-class Stata applications.) Also download data for the monthly risk-free return.

We selected ten value-weighted portfolios formed on operating profitability (OP) from Kenneth French's Data Library.

- Decile 1: lowest operating profitability.
- Decile 10: highest operating profitability.
- Risk-free return: the monthly RF column from the Fama/French three-factor dataset.
- Sample: July 1963–July 2026, covering 757 months.

2. Compute the monthly excess return for each decile portfolio.

For each decile and month:

$$\text{Excess Returns}_{d,t} = R_{d,t} - R_{f,t}$$

Here, $R_{d,t}$ is the decile's monthly return and is the $R_{f,t}$ same month's risk-free return.

Both inputs are reported in percent. We calculated this for all ten deciles across all 757 months.

3. Compute the sample mean and standard deviation for each decile portfolio excess return. For each decile portfolio, also compute the t -statistic and corresponding p -value for testing the null hypothesis that the mean excess return is zero. Report the results in a table. Discuss the results.

For each decile:

$$H_0: \mu_d = 0 \quad H_1: \mu_d \neq 0$$

The test statistic is:

$$t_d = \frac{x_d}{s_d/\sqrt{757}}$$

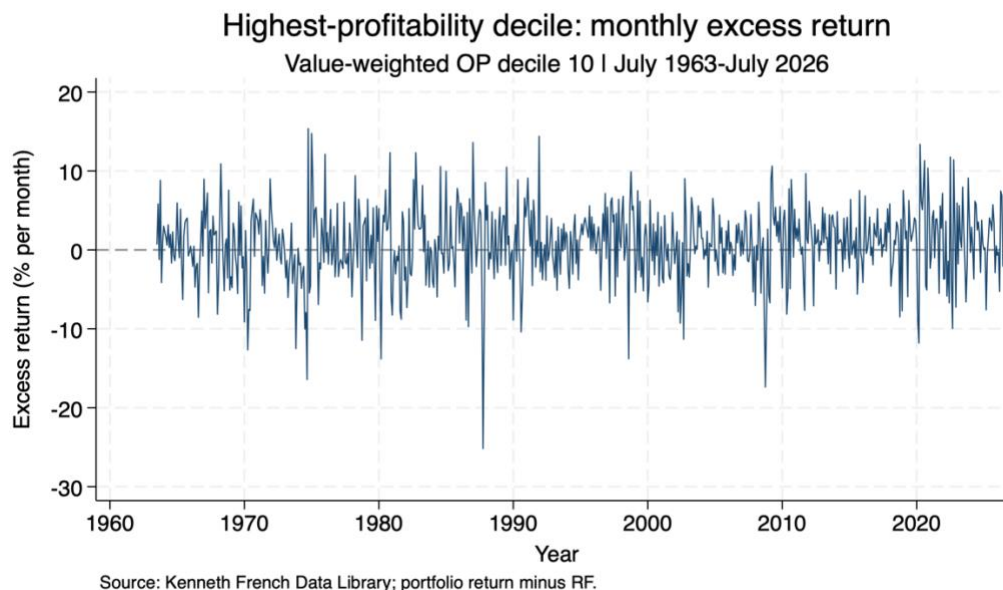
The two-sided p -value uses a Student's t distribution with **756 degrees of freedom**.

OP decile	Mean excess return (%)	Standard Deviation (%)	t-statistic	p-value
1	0.4394	6.5959	1.8327	1.8327
2	0.5142	5.3723	2.6336	0.0086206
3	0.4696	4.9263	2.6228	0.0088974
4	0.5619	4.6365	3.3345	0.0008961
5	0.6659	4.6271	3.9593	0.0000823
6	0.5924	4.5478	3.5837	0.0003605
7	0.5573	4.5699	3.3554	0.0008320
8	0.7272	4.6130	4.3375	0.0000164
9	0.7089	4.5306	4.3048	0.0000189
10	0.6910	4.6210	4.1143	0.0000431

All ten portfolios have positive sample mean excess returns. For deciles 2–10, we reject the hypothesis of zero mean excess return at the 5% level, and at 1%. For decile 1, the p-value is 0.0672, so we fail to reject at 5%, although we would reject at 10%.

Higher-profitability portfolios generally have higher average excess returns, but the relationship is not strictly increasing. Decile 8 has the highest mean, rather than decile 10. Decile 1 has the highest volatility, with a monthly standard deviation of 6.5959%.

4. Create a figure with a time-series plot for one of the decile portfolio excess returns.



The horizontal axis shows time, and the vertical axis shows monthly excess returns in percent. Returns fluctuate substantially despite their positive average. The largest negative excess return occurs in October 1987, at -25.17%.

5. Provide the Stata code in an appendix.

* Monthly excess returns for all ten deciles

forvalues d = 1/10 {

 generate double excess`d' = ret`d' - rf

}

* Means and sample standard deviations

tabstat excess1-excess10, statistics(n mean sd) format(%9.4f)

* Two-sided tests of zero mean excess return

forvalues d = 1/10 {

 ttest excess`d' == 0

}

* Time-series plot for the highest-profitability decile

twoway line excess10 month, ///

 title("Highest-profitability decile: monthly excess return") ///

 ytitle("Excess return (% per month)") ///

 xtitle("Year") ///

 yline(0)

graph export "output/op_decile10.png", replace